Electrochromic properties of perovskite $NdNiO_3$ thin films for smart windows

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Abstract: Thin films of perovskite nickelate $NdNiO_3$ (NNO) were deposited on single crystal ($LaAlO_3$ (LAO)) and polycrystalline (fluoride doped tin oxide coated glass (FTO)) substrates by magnetron sputtering, chemical solution deposition (CSD), and atomic layer deposition (ALD), respectively. Their electrochromic behaviors were investigated using a three-electrode setup in basic (KOH solution, pH=12) electrolyte. During bleaching/coloration process, the proton intercalation/de-intercalation and simultaneous electron compensation in NNO lattice under electrical bias led to crossover of the material between pristine conducting phase (Ni^{3+}) to

the strongly correlated insulating phase (Ni^{2+}), which serves as the working principle for electrochromic (tunable opacity) behavior. Cyclic voltammetry scans (CVs) demonstrates that NNO films are electrochemically stable in basic solutions, for all three film deposition methods explored here. CV scans at varying rates enabled extraction of diffusion coefficient of protons in thin film NNO, which is ~10^{-7} cm^2 s^{-1} among all films studied. Light transmittance modulation by bleaching and coloration was observed on films grown on both LAO and FTO substrates, suggesting its potential as an electrochromic material candidate for smart window and optical shutter applications. Porous NNO films obtained by chemical solution deposition tends to demonstrate stronger electrochromic activity than dense films grown by sputtering or ALD.

Keywords： Electrochromic; Perovskite; Nickelate; Thin film; Strongly correlated; Smart windows, Tunable optics

Introduction

Electrochromic materials change their opacity upon application of an electrical stimulus. The change in optical property can be non-volatile and opacity level tuned continuously by bias voltage. Such materials are actively being researched for smart windows via controlling transparency with potential applications in energy efficient building architectures, optical shutters, and tunable optical surfaces for eye goggles, aircraft or automotive vehicles.[1] Typical electrochromic materials operate in contact with a transparent electrolyte medium sandwiched between two electrodes. The electrolyte which can be in either liquid, gel, or solid form conducts mobile ions such as protons (H^+) and lithium ions (Li^+), [2, 3]. The fundamental operation principles of electrochromic inorganic oxides are based on the reversible injection of electrons and small-radius mobile cations into/out of a solid crystal lattice, accompanied with the electron compensation and resulting changes in

optical transparency. Time scales of interest in these diverse applications range from seconds to minutes wherein the optical transparency modulation range is controlled by electric bias over chosen time period.

To date, many inorganic oxide materials including WO_3,[4] niobium oxide (Nb_2O_5),[5] titanium oxide (TiO_2),[6] nickel oxide (NiO),[7] and iridium oxide (IrO_2),[8] whose color can be switched between colored (e.g. blue, grey or opaque) state and transmissive state have attracted much attention benefitted by their diverse composition/structure, superior electrochromic performance and stability again UV corrosion. [9]

The most widely studied model cathodic electrochromic system of WO_3 modulates its transmittance relying on the insertion of protons and charge-compensating electrons to fill into the conduction band (t_{2g}^1) of WO_3.[10]. The material undergoes an insulator-metal transition accompanied by variation in electrical resistivity and optical transmittance between dark blue and transparent (Figure 1a). The electrochromic properties can vary depending on the microstructure and synthesis methods. Zhao et al., synthesized quantum dots of WO_3 with an average size less than 2 nm, which showed coloration/bleaching time <1s and color efficiency of 154 cm^2 C^{-1}.[11] Cai et al., prepared porous WO_3 film which showed an ideal optical modulation capacity of 97.7% at wavelength of 632.8 nm with switching times of 6 and 2.7 s for the coloration and bleaching, respectively. [12] Compared to crystalline WO_3 film, amorphous WO_3 may show quicker coloration and improved efficiency. Cheng *et. al.* reported an amorphous WO_3 prepared by photo-deposition with an optical modulation of 70% at 700 nm and coloration efficiency > 130 cm^2/C^{-1}.[13] Microstructure and synthesis methods are therefore an important factor in optimizing performance of electrochromic materials and an active area of research.

In comparison, the most commonly used anodic oxide-based EC materials of Ni oxides are able to change from a transparent state to a neutral colored one upon extraction of protons or insertion of OH^- ions. Charge-balancing electrons are simultaneously extracted from the valence band to convert Ni^{2+} to Ni^{3+}. During

operation, the films are actually in the state of mixed oxide and hydroxide at bleached state, and of oxygen-rich compounds NiOOH and Ni_2O_3 at colored state which strongly absorb visible light. [14] Similarly, NiO with diverse nanostructures such as nano-rods, nano-sheet and nano-flake have been reported.[15] Ultrathin NiO nano-sheet was reported to have a transmittance modulation of 40% with a coloration efficiency of 43.5 $cm^2 C^{-1}$.[16] Patil et al. reported one dimensional (1D) NiO nano-rods with significantly enhanced optical transmittance capability of 60%, and coloration and bleaching times of 1.55 and 1.22 s at 630 nm, respectively. Ma et al. demonstrated the synthesis of hierarchical NiO with nano-flake morphology with high optical modulation (73.2% at 550 nm), and switching time of 1.8s for coloration and 3.2s for bleaching. [17]

The ionic diffusion co-efficient in the switching medium places fundamental limits on operational characteristics of electrochromic materials. For example, crystalline WO_3 and NiO have typical proton diffusivity of the order of $10^{-8} \sim 10^{-10}$ cm^2 s^{-1}.[18, 19] Therefore, new materials that offer comparable or improved ionic diffusion coefficient and absorption over the visible spectrum are of interest and worth exploring as potential candidates for next generation electrochromic windows.

Rare earth nickelates with perovskite structure (ABO_3) are of broad research interest due to its highly sensitive electronic phase diagram with respect to orbital electron occupancy.[20-24] As a representative, $NdNiO_3$ (NNO) shows metallic conduction behavior and is neutral grey at room temperature with single electron occupancy in e_g orbital of Ni_{3d} bands (e_g^1).[25-29] Upon cation intercalation accompanied with extra electron filling, the Ni state in pristine NNO experiences crossover to Ni^{2+} state that is highly localized ((e_g^2)).[30, 31] The strong carrier localization opens up an optical gap and makes the material transmissive. The ion-electron intercalation therefore leads to completely opposing effects compared with WO_3 as shown schematically in figure 1.

While hydrogen doping-driven changes in conductivity of $NdNiO_3$ have been reported in literature in the context of memory device technologies[24, 26, 32], properties relevant to optical window applications are still largely unknown and forms the subject of this study. Here, we present experimental results on nickelate films that display promising characteristics pertaining to optical transmittance and we further compare its properties with well-studied oxide counterparts such as WO_3 and NiO.

Experimental:

Physical vapor deposition (PVD) of $NdNiO_3$ thin films (PVD-NNO): $NdNiO_3$ (NNO) thin films were grown on (001) $LaAlO_3$ (LAO) and fluoride doped tin oxide (FTO, thickness 180 nm) coated glass substrates using magnetron sputtering at room temperature combined with post-annealing in air atmosphere. The deposition condition is 40/10 sccm Ar/O_2 mixture at total pressure of 5 mTorr from two metallic Ni (DC, 70W) and Nd (RF, 170W) targets. The deposition rate is 140 nm per hour. The as-deposited samples were annealed in air at 500 °C for 24 h in a tube furnace to form the perovskite phase. Unless specifically noted, film thickness is ~50 nm prepared for the characterization studies.

Chemical solution deposition (CSD) of $NdNiO_3$ thin films (CSD-NNO): The solution was prepared by dissolving neodymium nitrate hexahydrate ($Nd(NO_3)_3{\cdot}6H_2O$, sigma, purity 99.99%) and nickel(II) acetate tetrahydrate ($Ni(CH_3COO)_2{\cdot}4H_2O$, sigma, purity >99.0%) in the solvent mixture of 2-methoxyethanol and acetic acid (volume ratio 4:1), preheated at 80°C. The solution was heated and stirred at 80°C for 1h and then filtered. The concentration of the solution was controlled to 0.2M by properly weighing the two solid precursors. The spin-coating method was used to deposit NNO thin films with spin speed of 3000rpm and duration of 30s. Then the coated films were baked at 175°C for 20min to evaporate any remaining solvent or water molecules. Then the baked film was immediately transferred to the furnace for pyrolysis and annealing.

100 The pyrolysis condition was 300°C for 30 min in air. Subsequently, the films were annealed at 600°C in air for
101 12h. The film thickness was ~40 nm on LAO and ~100nm on FTO glass substrate.

102 **Atomic layer deposition (ALD) of $NdNiO_3$ thin films (ALD-NNO):** ALD depositions of $NdNiO_3$ were
103 performed in a Veeco Savannah S200 reactor. The deposition temperature was kept at 160 °C and pressure was
104 0.9 mbar, maintained by a 20 SCCM of N_2 (Indiana Oxygen, 99.9995%). $Nd(iPrCp)_3$ (iPrCp =
105 i-propylcyclopentadienyl, Strem Chemicals, 99.9%) and $Ni(tBu2\text{-}amd)_2$ (tBu2-amd =
106 N,N-di-t-butylacetamidinato, Strem Chemicals, 99.999%) were used as organometallic precursors and
107 evaporated at 160 °C and 125 °C, respectively, while ozone (O_3) with ~concentration 200ng/mL was
108 employed as the oxidant. $Nd(iPrCp)_3$ was introduced into the reactor using a "boosted" configuration where N_2
109 was pressurized into the cylinder prior to dosing precursor into the reactor. Ozone (O_3) was used as an oxygen
110 precursor. Pulse and purge durations were 0.5/5 s for $Nd(iPrCp)_3$, 0.75/5 s for $Ni(tBu2\text{-}amd)_2$, and 0.5/5 s for
111 O_3. The $NdNiO_3$ film were deposited on single crystalline (001) $LaAlO_3$ (LAO), indium doped tin oxide (ITO)
112 coated glass substrates, and fluorine doped tin oxide (FTO) coated glass substrates (MTI Corp.). All substrates
113 were washed with toluene, acetone, and isopropanol and dried with Ar gas. After the deposition, the films were
114 annealed at 600 °C for 24 hours in air for formation of perovskite phase. Extensive optimization experiments
115 with precursors over a two-year period were carried out as this is the first report on synthesis of ALD-grown
116 $NdNiO_3$ films. Complete details of the ALD process optimization characteristics are included in the
117 Supplement file.

118 **Electrochemical measurements:** 0.01M (pH=12) basic (KOH) aqueous solution was utilized to study the
119 electrochromic properties of NNO in basic aqueous environment. A three-terminal electrochemical cell were
120 used for both electrochromic and cyclic voltammetry (CV) measurement. A silver (Ag) wire was bonded on

the thin films with silver paste. The wire contact of the electrode was covered with epoxy such that the influence of Ag wire can be excluded. The NNO film was then submerged into aqueous solutions and denoted as a working electrode. The counter-electrode was a graphite rod with large surface area. Ag/AgCl (saturated KCl) was used a reference electrode. For a specific cyclic voltammetry measurement, a static electric potential (vs. Ag/AgCl) (10, 20, 50 and 100 mV s^{-1}, respectively) was applied to NNO film by a potentiostat using a Solartron 1260A electrochemical analyser. To obtain the diffusion coefficient of proton in NNO film, Randles-Sevcik (R-S) equation (i.e., $I_p/A = 268600\ n^{1.5}C(Dv)^{0.5}$) was used to fit the relation between peak cathodic current density (I_p/A, in amps/cm^2) and square root of scan rate ($v^{0.5}$), where C is the concentration of reaction-involved species in solution (mol/mL); D is the diffusion coefficient of proton in NNO at room temperature ($cm^2\ s^{-1}$); v is scan rate (V s^{-1}); n is the number of electrons exchange in the redox process between Ni^{3+} and Ni^{2+} which is 1.[33, 34]

X-ray diffraction: X-ray diffraction (XRD) patterns were collected using Panalytical MRD X'Pert Pro XRD and Bruker D8 focus XRD instruments.

Raman Spectroscopy: Raman spectra were measured using an inVia confocal Raman micro-spectrometer (Renishaw, UK), utilizing a 532 nm solid state laser.

Transmittance modulation of films: Optical characterization was carried using a UV−vis−NIR spectrophotometer (PerkinElmer Lambda 1050 and Lambda 950). The bare substrate (LAO or FTO coated glass) was used as a reference for each measurement.

Scanning electron microscope (SEM): SEM images of the NNO films were taken by Thermo Scientific™ Apreo SEM and Hitachi S-4800 field emission SEM (FE-SEM).

Transmission electron microscope (TEM): A cross-section sample was lifted out of NNO samples prepared by CSD and ALD by a Zeiss 1540XB CrossBeam focused ion beam FE-SEM system. Using a Gatan precision ion polishing system (PIPS) II these samples were further thinned down to less than 100 nm. The cross-sectional sample of PVD-NNO was prepared by conventional polishing. High-angle annular dark-field scanning transmission electron microscope (HAADF-STEM) imaging and X-ray energy-dispersive spectroscopy (EDS) mapping were performed with a FEI Talos F200X TEM. High-resolution transmission electron microscopy (HR-TEM) characterizations on representative samples grown by PVD was perform by a FEI Talos F200X TEM operated at 200kV, while the films grown by CSD and ALD were performed using the Argonne chromatic aberration-corrected TEM (a FEI Titan 80-300 ST with an image aberration corrector to correct both spherical and chromatic aberrations) at an accelerating voltage of 200 kV.

Results

1. Electrochromic behaviors of PVD-NNO, CSD-NNO and ALD-NNO

NNO thin films with nominal thickness of 40~50nm were deposited onto single crystal $LaAlO_3$ (LAO) substrates by PVD, CSD and ALD respectively. Figure 2(a-c) shows theta-2theta XRD patterns of NNO grown on LAO substrates and the insets show the enlarged patterns at 2θ=45-50°. The diffraction pattern indicates the growth of the NNO film deposited by each method is highly [001] oriented ($2\theta_{LAO}$ at 47.98°, $2\theta_{PVD\text{-}NNO}$ at 47.47°, $2\theta_{CSD\text{-}NNO}$ at 47.48° and $2\theta_{ALD\text{-}NNO}$ at 47.54°) without any contribution from secondary phases.[35] Such an epitaxial relation can be understood based on the close matching of the pseudocubic unit cell parameter of NNO (a_{NNO}=3.81 Å) with the cubic unit cell parameter of LAO (a_{LAO}= 3.789 Å) and has been noted in literature.[36]

The morphology of representative epitaxial NNO/LAO films were analyzed by transmission electron microscopy (TEM) and shown in Figure 2(d-i). The high-angle annular dark field (HAADF-STEM) image and energy dispersive X-ray (EDX) mapping (Figure 2(d-f)) results demonstrate the uniform dispersion of Nd and Ni and the estimated atomic ratio is ~1 across the film thickness, for the three films. Compared with PVD-NNO/LAO and ALD-NNO/LAO, CSD-NNO/LAO film was relatively porous due to the nature of solution deposition followed by high temperature annealing that removes the binder [37-40]. Cross-section high resolution TEM images of NNO/LAO films are shown in Figure 2(g-i), suggesting the oriented growth of NNO by all three deposition methods.

Raman spectra evolution of NNO/LAO films before and after proton uptake are shown in Figure 3. The Raman peaks of LAO substrate between 450 and 550 cm^{-1} could be identified in all spectra. Generally, the distorted perovskite nickelate exhibits 24 Raman-active modes but only a few of them can be detected in experiment.[41] The pristine NNO films show the expected peaks at ~300cm^{-1} (E_g mode), ~410cm^{-1} (T_{2g} mode) and ~438cm^{-1} (T_{2g} mode) which is consistent among all three samples.[41, 42] All peak intensity from NNO, especially the T_{2g} peak at ~436cm^{-1}, strongly decreased after the films were bleached in the basic electrolyte and later partially recovered after coloration treatment. The bleaching conditions for PVD-NNO/LAO and ALD-NNO/LAO samples were -0.7V vs. Ag/AgCl for 120s while for CSD-NNO/LAO it was -0.7V vs. Ag/AgCl for 60s. The coloration conditions for PVD-NNO/LAO and ALD-NNO/LAO films were +1V vs. Ag/AgCl for 180s, and +1V vs. Ag/AgCl for 120s for CSD-NNO/LAO sample.

Cyclic voltammetry scans (CVs) were performed by a three-electrode electrochemical cell consisting of a working electrode (NNO films), a counter electrode (graphite rod), and a reference electrode (Ag/AgCl). The CVs were measured on NNO/LAO films in 0.01M KOH (pH=12) electrolyte with a linear potential sweep at

various scan rates ranging from 100mV/s to 10 mV/s to identify the insertion/extraction process of protons with a visible color change. Pairs of redox peaks (at -0.6V/-0.1V) can be observed on all three NNO/LAO films during the charge/discharge process, which indicates that the capacity behavior in CV measurement is due to Faradaic redox reactions between Ni^{3+} and Ni^{2+}. The electrochromic phenomena of NNO in 0.01M KOH solution could be described as follows:

$$NdNiO_3(opaque) + H_2O + e^- \underset{coloration}{\overset{bleaching}{\rightleftharpoons}} H\text{-}NdNiO_3(transparent) + OH^-$$

During CV measurement, specifically, the anodic current peaks at high potentials correspond to the charge transfer as the Ni^{2+} is oxidized. This peak shifted to low-potential regime as the scan rate decreased, indicating that the reaction is kinetically limited by the charge and mass transfer. Figure 4(d-f) shows the linear fitting between maximum peak anodic current density (I_p/A) and the square root of the scan rate ($v^{0.5}$) by using the R-S equation.[33] Based on the extrapolated slopes, the estimated proton diffusion coefficients of PVD-NNO/LAO, CSD-NNO/LAO and ALD-NNO/LAO was 1.14×10^{-7} cm^2 s^{-1}, 1.16×10^{-7} cm^2 s^{-1} and 1.3×10^{-7} cm^2 s^{-1}, respectively. A comparison of proton diffusion coefficient with other electrochromic metal oxides is shown in Table I. In comparison, proton diffusion coefficient for WO_3 ranging between ~2×10^{-7} and 4×10^{-10} cm^2 s^{-1} and in NiO ranging from 10^{-10} to 10^{-12} cm^2 s^{-1} has been reported in literature, which is both crystallinity and electrolyte dependent.[18, 19, 43-45]

The electrochemical stability of NNO films in pH=12 environment was investigated by continuous cycling of the applied potential from −0.7 V to +0.7 V (vs. Ag/AgCl) at 100 mV s^{-1}. The changes of the film electroactivity, which is represented by the charge capacity within each CV cycle, are used to evaluate the stability.[46-48] It was observed from the electrochemical stability graphs that PVD-NNO/LAO, CSD-NNO/LAO and ALD-NNO/LAO can maintain >85% of their initial electroactivities after 400 cycles (Figure 5). This result

shows that despite the choice of deposition method, NNO/LAO films consistently show a good electrochemical stability in basic solution. This observation is simlar to previous works on binary NiO stability.[10]

Figures 6(a-c) shows the optical transmittance spectra of NNO/LAO samples in their pristine (no treatment), bleached and colored states measured in the visible wavelength range of 350–800 nm. Optical transmittance spectrum from a bare LAO substrate was used as the reference. At the wavelength of 632.8 nm, the optical transmittance modulation of NNO films are listed in Table II. The bleaching treatment (-0.7V vs. Ag/AgCl for 30s) led to the transmittance increase of PVD-NNO/LAO, CSD-NNO/LAO and ALD-NNO/LAO by 24%, 38% and 15%, respectively. The transmittance of CSD-NNO after bleaching was higher than 90%. Further coloration treatment modulates the transmittance of PVD-NNO, CSD-NNO and ALD-NNO films by 14%, 24% and 10%, respectively. Due to the larger transmittance change in CSD-NNO after bleaching, the coloration time was slightly longer than that for PVD-NNO and ALD-NNO samples.

The coloration efficiency (CE) is defined as the change in optical density (ΔOD) per injected charge density (Q) at a specific wavelength (632.8 nm in our case). The change in optical density is obtained from equation (3-4):

$$\mathrm{CE} = \frac{\Delta OD(\lambda)}{Q} \quad (3)$$

$$\Delta OD(\lambda) = \log\left(\frac{T_b}{T_c}\right) \quad (4)$$

Where T_c and T_b are the transmittance in the colored and bleached states, respectively. The relationship between ΔOD and the charge density variation were shown in Figure 6(d-f). CE can be estimated from the slope of the linear fitting range of the curve. This method is commonly used for studying electrochromic materials[37, 49-53]. The coloration efficiency of dense films like PVD-NNO/LAO or ALD-NNO/LAO is less than

20 cm^2 C^{-1}, as shown in figure 6 (d) and (f). The coloration efficiency for CSD-NNO/LAO could reach 47.9 $C^{-1}cm^2$ (figure 6(e)). It is close to the reported values from porous WO_3 and NiO (typically around 40~50 cm^2 C^{-1} [19, 51, 54]), and is 2-3 × the value obtained for PVD-NNO/LAO or ALD-NNO/LAO thin films. A primary reason for the difference among NNO films deposited by different methods could be the denser nature of the PVD and ALD films that limits the interfacial area between the active metal oxide grains and the protons from electrolyte, limiting the electrochromic efficiency. [55, 56] The SEM images in Figure 6(g-i) confirms that CSD-NNO/LAO film has a relatively porous, granular morphology and could be suitable for further optimization of electrochromic activity.

Finally, we examined the properties of NNO films grown by the above methods on FTO coated glass substrate for potential smart windows application. Figure 7(a-c) shows XRD patterns of the NNO film grown on FTO coated glass. Besides the diffraction peaks of FTO, three XRD diffraction peaks at 33.3°, 47.78°, and 59.5° belonging to polycrystalline NNO were observed[57]. The electrochemical stability of NNO/FTO films were also examined by cyclic voltammetry, which confirms that polycrystalline NNO could maintain >80% of the initial activity after 400 cycles (Figure 8(a-f)). We noticed the peak-to-peak separation in PVD-NNO/FTO and ALD-NNO/FTO are larger than that of corresponding NNO films grown on LAO substrates. This could be likely due to the limited diffusion as well as charge transfer in dense films deposited by PVD and ALD, as shown in Figure 7(d)(f). However, CSD-NNO/FTO showed a similar CV peak potentials to single crytalline NNO/LAO films due to the microstructural differences as discussed in the above section. The transmittance change between colored and bleached states of NNO/FTO films are shown in Figure 8(g-i). At the wavelength of 632.8 nm, the transmittance modulation of NNO/FTO samples are listed in Table II (transmittance of a bare FTO coated glass substrate was used as the reference). The bleaching condition (-0.7V vs. Ag/AgCl for 30s)

led to the transmittance enhancement of PVD-NNO, CSD-NNO and ALD-NNO, all with a thickness of ~50nm, by 27%, 24% and 11%, respectively. Note the bleaching response of CSD-NNO/FTO is limited due to the high transmittance in the pristine state (67%). The inset of Figure 8 (h) shows the transmittance modulation of 100nm thick CSD-NNO film grown on FTO, for which the pristine transmittance at 632.8nm (55%) is comparable to that of 50nm PVD-NNO/FTO (48%) or 50nm ALD-NNO/FTO (61%). After bleaching (-0.7V vs. Ag/AgCl for 30s) the tranmittance of 100nm CSD-NNO/FTO was increased by ~40%. Further coloration treatment could modulate the tranmittance of each sample back to opaque state. This result is similar to the transmittance modulation trend observed in NNO films grown on LAO substrates. Optical images of the transmittance change of NNO/FTO galss could be found in Figure S2.

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2. Comparison of film performance between synthesis methods

NNO thin films prepared by physical vapor deposition (PVD), chemical solution deposition (CSD) and atomic layer deposition (ALD) grew in an oriented manner on LAO substrates and were polycrystalline on FTO glass substrates (Figures 2 and 7(a-c)). The films deposited in each method exhibited reversably tunable optical transmittance due to proton (/ de-) intercalation (Figures 3, 6(a-c), and 7(g-i)), with a proton diffusion coefficient of ~$1*10^{-7}$$cm^2$/s (Figures 4 (d-f)). Also, regardless of the deposition method, all films were stable in pH=12 basic solution, and could maintain more than 80% initial electroactivity after 400 cyclic votammetry scans (Figures 5 and 7(d-f)). Due to the porous nature of chemical solution deposited NNO films, they are more sensitive to bleaching and coloration treatments than dense NNO films deposited by PVD or ALD. For the same bleaching condition (-0.7V vs. Ag/AgCl for 30s), CSD-NNO/LAO or NNO/FTO increased transmittance by ~40% (Table II) at 632.8nm, while this value for PVD-NNO/LAO or ALD-NNO/FTO was ~25% (Table II). The transmittance of bleached CSD-NNO (either deposited on LAO or FTO glass) film is

higher than 90%. After bleaching, the coloration efficiency for CSD-NNO ($47.9 C^{-1} cm^2$) is also much higher than that of PVD-NNO ($17.9 C^{-1} cm^2$) and ALD-NNO ($19.8 C^{-1} cm^2$) (Figure 6(d-f)). For application in smart windows, CSD-NNO films appears to show greater degree of optical tunability compared to its denser counterparts synthesized by PVD or ALD. Similar enhancement of electrochromic performance due to microstructural aspects such as porosity has been noted in other oxides as well in literature.[7, 14, 49, 51]

Conclusions

We successfully synthesized a series of perovskite nickelate of $NdNiO_3$ (NNO) thin films on $LaAlO_3$ and FTO-coated glass substrates by PVD, CSD and ALD methods. Their electrochromic properties in basic electrolyte (0.01M KOH, pH=12) were systematically studied. The films demonstrated tunable opacity at the visible wavelength via insertion/extraction of protons from aqueous electrolytes. Cycling experiments indicated the material is stable in basic electrolyte. The high proton diffusion coefficient in NNO coupled with electrochemical stability is promising towards further exploration in smart window applications. Future directions to consider include further optimization of microstructural features during deposition to enhance kinetics of chromic tuning.

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Abbreviations:

PVD-NNO: neodymium nickelate ($NdNiO_3$) thin film grown by physical vapor deposition (sputtering); CSD-NNO: neodymium nickelate ($NdNiO_3$) thin film grown by chemical solution deposition (spin-coating); ALD-NNO: neodymium nickelate ($NdNiO_3$) thin film grown by atomic layer deposition; LAO: lanthanum aluminate (LaAlO3) substrate; FTO glass: fluorine doped tin oxide coated glass substrates.

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448 **Table I.** Diffusion coefficient of protons in representative electrochromic oxides

Materials	Electrolyte	R.T. Proton diffusion coefficient ($cm^2\ s^{-1}$) [a]	Calculation Method	Preparation Method	Reference
Polycrystalline WO_3	1N H_2SO_4	4×10^{-8}	R-S equation	Evaporation	[44]
Crystalline WO_3	Poly-AMPS[b]	2×10^{-11}	AC impedance	Evaporation	[43]
Amorphous WO_3	Poly-AMPS[b]	4×10^{-10}	AC impedance	Evaporation	[43]
Amorphous WO_3	Water-based electrolyte	$2.5 \times 10^{-7} \sim 2.5 \times 10^{-9}$	N/A	N/A	[58]
Amorphous WO_3	Organic molecules	$\leq 2.5 \times 10^{-13}$	Fick's Law	N/A	[19]
Crystalline WO_3	Organic molecules	$\geq 2.5 \times 10^{-7}$	Fick's Law	N/A	[19]
$Ni_{1-x}V_xO_y$	1M KOH	2.5×10^{-16}	R-S equation	Sputtering	[18]
NiOH	1M $HClO_4$	4.0×10^{-11}	Finite Diffusion Model	Commercial	[45]
MoO_3	1M H_2SO_4	1.55×10^{-10}	R-S equation	CVD	[59]
TiO_2	1 M $(NH_4)_2SO_4$	2×10^{-11}	R-S equation	PVD	[60]
Single crystalline NNO by PVD	pH=12 KOH	1.14×10^{-7}	R-S equation	PVD	*This work*
Single crystalline NNO by CSD	pH=12 KOH	1.16×10^{-7}	R-S equation	CSD	*This work*
Single crystalline NNO by ALD	pH=12 KOH	1.30×10^{-7}	R-S equation	ALD	*This work*

449 [a.] R.T.: room temperature

450 [b.] Polymerized 2-acrylamido-2-methylpropane sulfonic acid

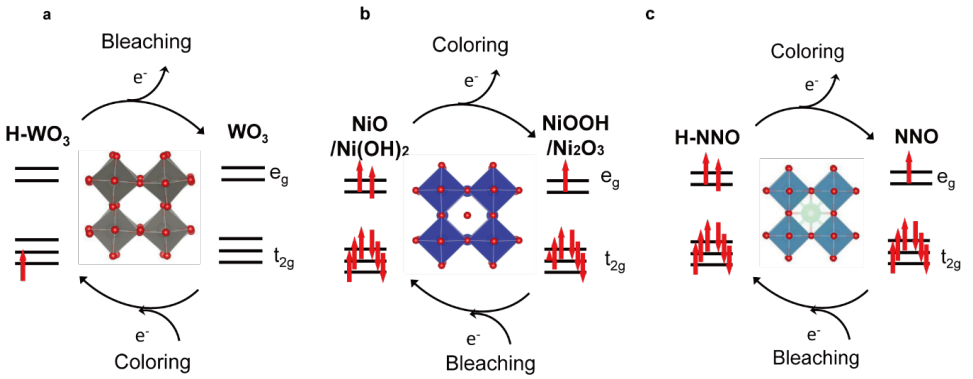
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452 **Table II.** Optical transmittance modulation of NNO films at 632.8nm

Materials	Pristine	Bleached (-0.7V vs. Ag/AgCl 30s)	Colored (+1V vs. Ag/AgCl 60s)
50nm PVD-NNO/LAO	50%	74%	60%
50nm CSD-NNO/LAO	56%	94%	70% [a]
50nm ALD-NNO/LAO	57%	72%	62%
50nm PVD-NNO/FTO glass	48%	75%	58%
50nm CSD-NNO/FTO glass	67%	91%	74%
100nm CSD-NNO/FTO glass	55%	94%	60% [a]
50nm ALD-NNO/FTO glass	61%	72%	67%

453 [a.] The coloration condition was +1V vs. Ag/AgCl for 180s

454



455

456 **Figure 1. Schematic diagram of working principle of various electrochromic materials. (a)** WO_3 working

457 principle. The grey and red colors represent W and O atoms, respectively. The pristine WO_3 is optically

458 transmissive (bleached state) and becomes opaque after proton insertion accompanied with extra electron

459 filling into t_{2g} orbital (colored state). **(b)** NiO working principle. The dark blue and red colors represent Ni and

460 O atoms, respectively. The NiO is naturally transparent at pristine state and colored while intercalated with

461 extra OH^- or further oxidized to Ni_2O_3. **(c)** $NdNiO_3$ working principle. The green, light blue and red colors

462 represent Nd, Ni and O atoms, respectively. The nickelate is naturally opaque with single electron occupancy

463 in e_g orbital and experiences a crossover to double occupancy state after proton intercalation becoming

464 transmissive.

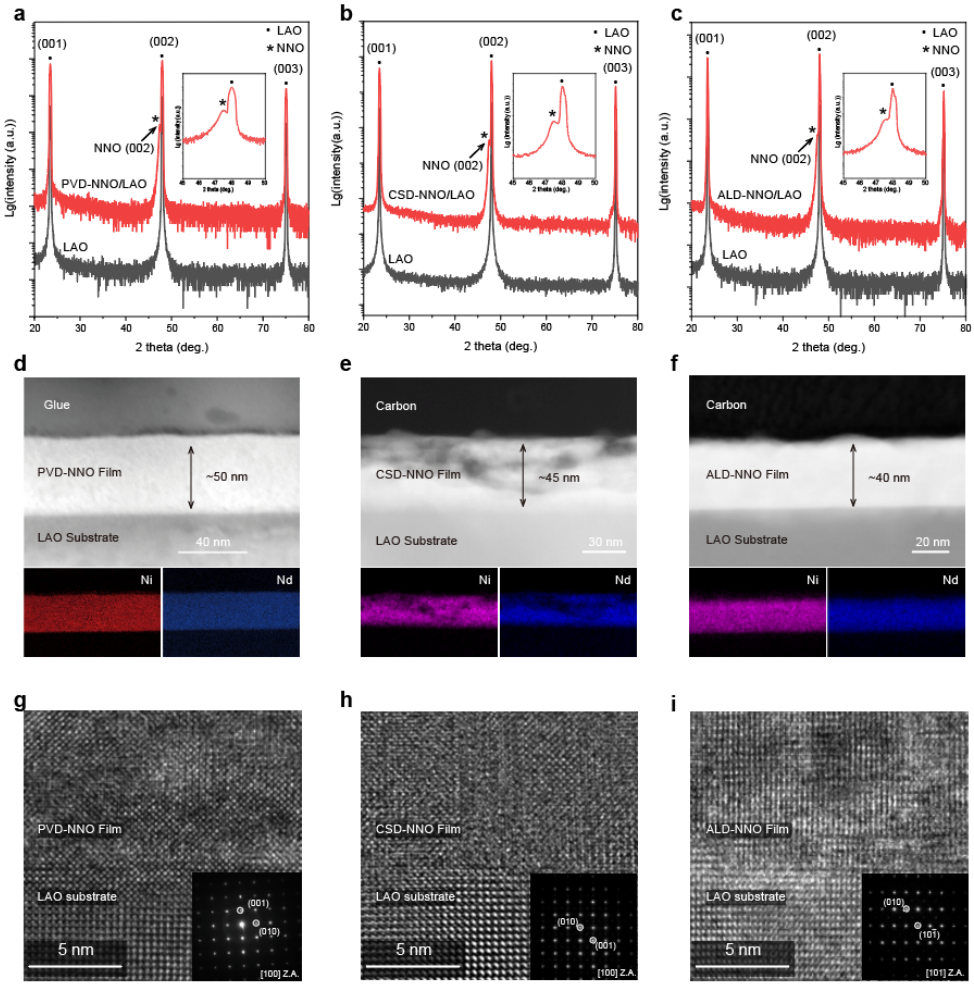
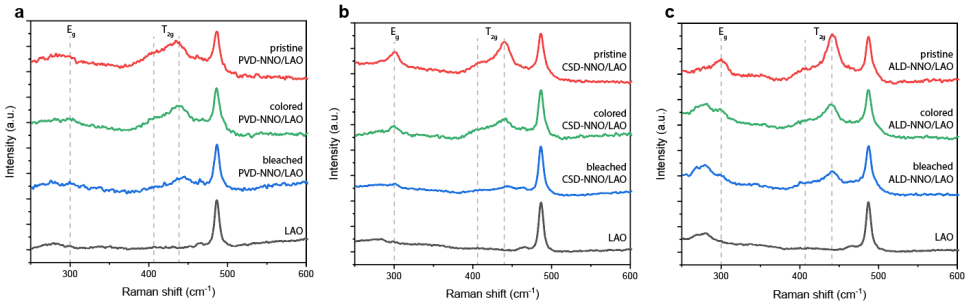


Figure 2. NNO thin films on LAO substrate. (a-c) XRD scans of NNO/LAO samples deposited by PVD, CSD and ALD. The peaks are labeled with pseudocubic notation. Insets show the enlarged scan from 45° to 50°. NNO (002) peaks were all identified, with $2\theta_{PVD\text{-}NNO}$ at 47.47°, $2\theta_{CSD\text{-}NNO}$ at 47.48° and $2\theta_{ALD\text{-}NNO}$ at 47.54°, which indicates our NNO films on LAO substrates are oriented. No peaks from any secondary phases were observed. **(d-f)** Cross-sectional HAADF-STEM and the corresponding EDX mappings of Ni and Nd

elements, which indicate a 1:1 atomic ratio in all films. The CSD-NNO/LAO thin film is porous while the other two are dense. **(g-i)** High resolution TEM images of cross-sectional NNO/LAO samples at the film/substrate interface. The selected area diffraction patterns are shown in the insets, also labeled with pseudocubic notation.



475

476 **Figure 3. Raman spectra of NNO/LAO samples at pristine, bleached and colored states.** Vibration modes
477 from pristine NNO film noted in literature are labeled by the dashed lines.[41, 42] The NNO peak intensities
478 decreased after bleaching (proton intercalation step) and partially recovered after coloration (proton
479 de-intercalation step). **(a)** NNO/LAO sample prepared by PVD. Bleaching condition: -0.7V vs. Ag/AgCl for
480 120s. Coloration condition: +1V vs. Ag/AgCl for 180s. **(b)** NNO/LAO sample prepared by CSD. Bleaching
481 condition: -0.7V vs. Ag/AgCl for 60s. Coloration condition: +1V vs. Ag/AgCl for 120s. **(c)** NNO/LAO sample
482 prepared by ALD. Bleaching condition: -0.7V vs. Ag/AgCl for 120s. Coloration condition: +1V vs. Ag/AgCl
483 for 180s. The films at bleached and colored states were treated in pH=12 electrolyte using a three-electrode
484 setup.

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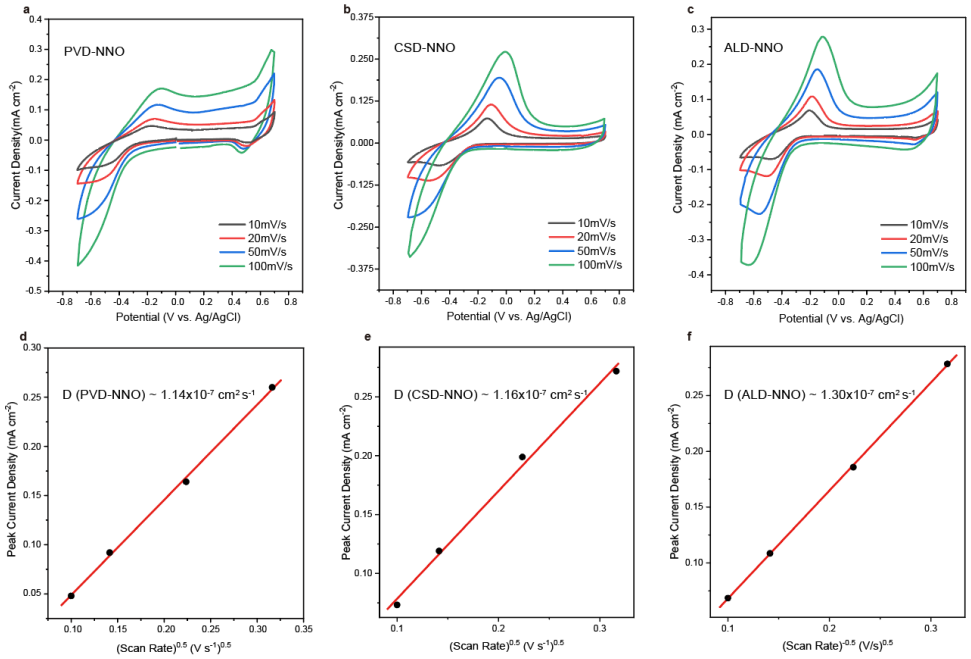
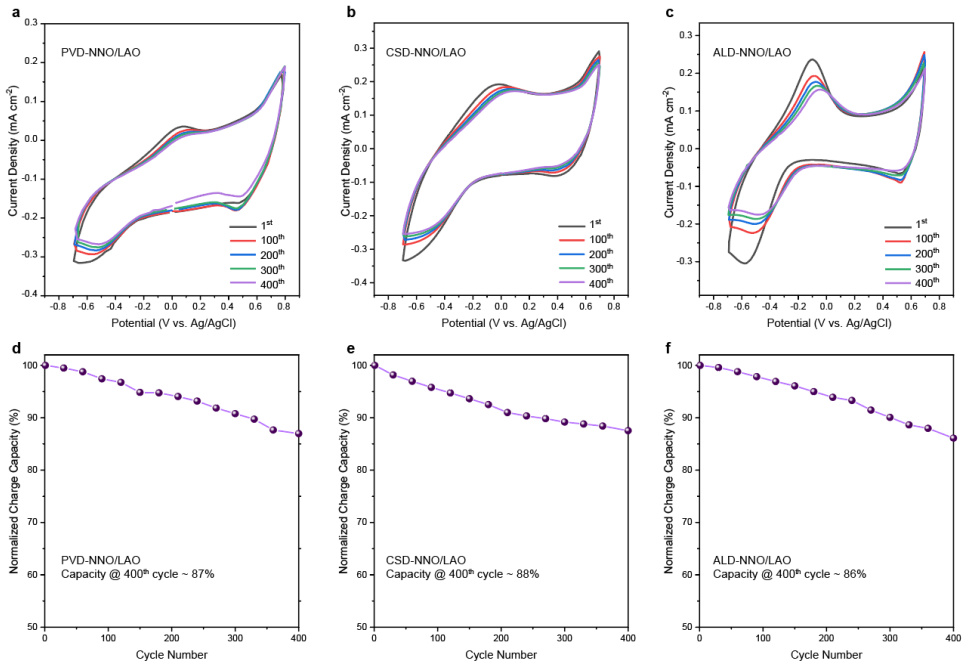


Figure 4. CV scans at various scan rates and the corresponding proton diffusion coefficients in NNO/LAO films. NNO films deposited by **(a)** PVD, **(b)** CSD and **(c)** ALD on LAO substrate were scanned by CV technique at 10mV/s, 20mV/s, 50mV/s and 100mV/s in pH=12 electrolyte. **(d-f)** Linear fittings between peak anodic current density (I_p/A) and the square root of the CV scan rate ($v^{0.5}$) were used to extract the proton diffusion coefficient in films, D, which are noted in the plots. In all cases, the proton diffusion coefficient was extracted to be in the similar range of ~10^{-7} cm^2 s^{-1}.



494

495 **Figure 5. Electrochemical stability of various NNO/LAO samples in pH=12 electrolyte. (a)**
496 PVD-NNO/LAO, **(b)** CSD-NNO/LAO and **(c)** ALD-NNO/LAO were examined by CV scans with a scan rate
497 of 100mV/s for 400 cycles. The initial charge capacity after 400 CV cycles for **(d)** PVD-NNO/LAO, **(e)**
498 CSD-NNO/LAO and **(f)** ALD-NNO/LAO was ~87%, 88% and 86%, respectively, which indicates the
499 NNO/LAO samples are quite stable in basic solution.

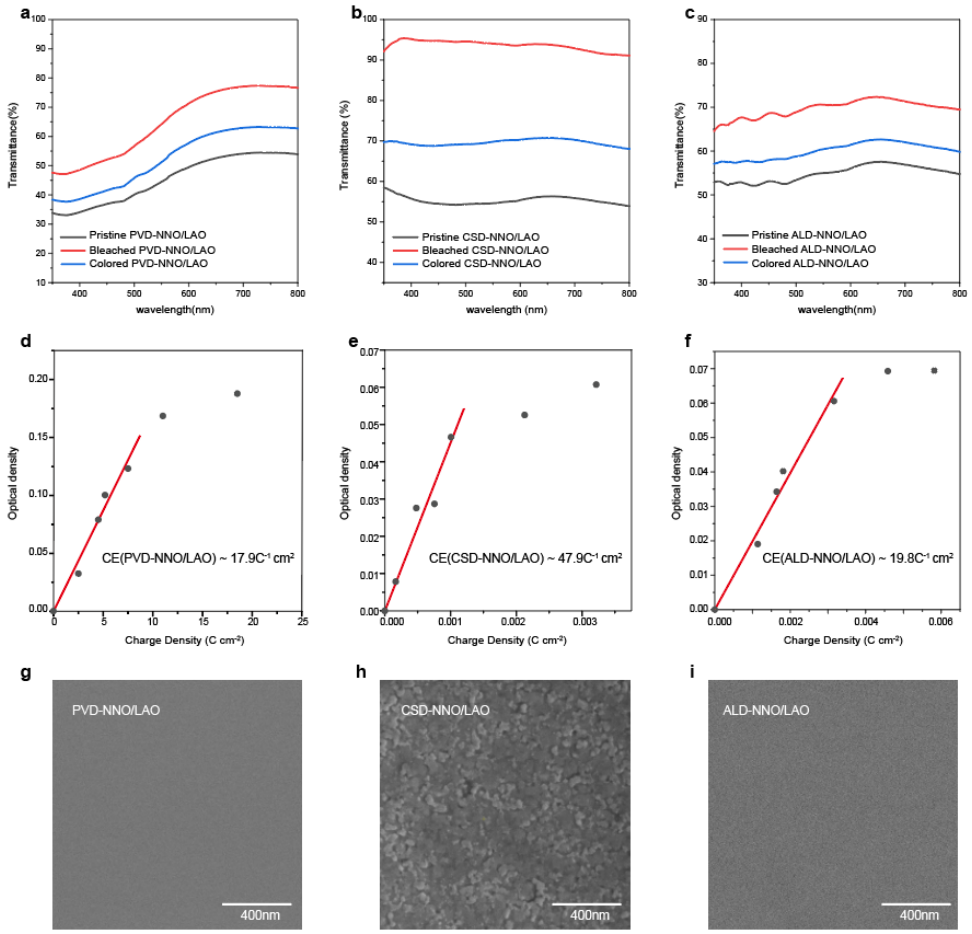


Figure 6. The capability of optical transmittance modulation on ~50nm NNO/LAO films and the extracted coloration efficiency. (a-c) UV-Vis spectra of NNO/LAO films prepared by **(a)** PVD, **(b)** CSD and **(c)** ALD in their pristine (no treatment), bleached (-0.7V vs. Ag/AgCl 30s) and colored (+1V vs. Ag/AgCl 60s for **(a)** and **(c)** and 180s for **(b)** due to the dramatic transmittance increase of CSD-NNO after bleaching) states in pH=12 electrolyte. **(d-f)** Coloration efficiency (CE) of NNO/LAO films prepared by **(d)** PVD, **(e)** CSD and **(f)** ALD. By linear fitting, the extracted CE values are noted in each figure. The SEM images of **(g)**

PVD-NNO/LAO and **(i)** ALD-NNO/LAO indicates the films prepared by these two methods are dense, while SEM of **(h)** CSD-NNO/LAO shows the film is relatively porous and granular.

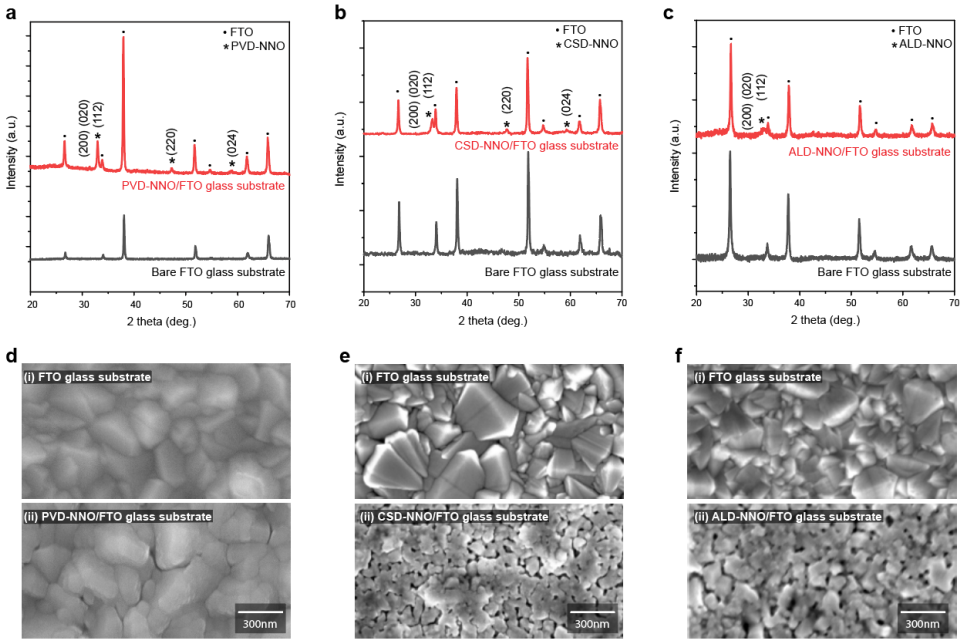
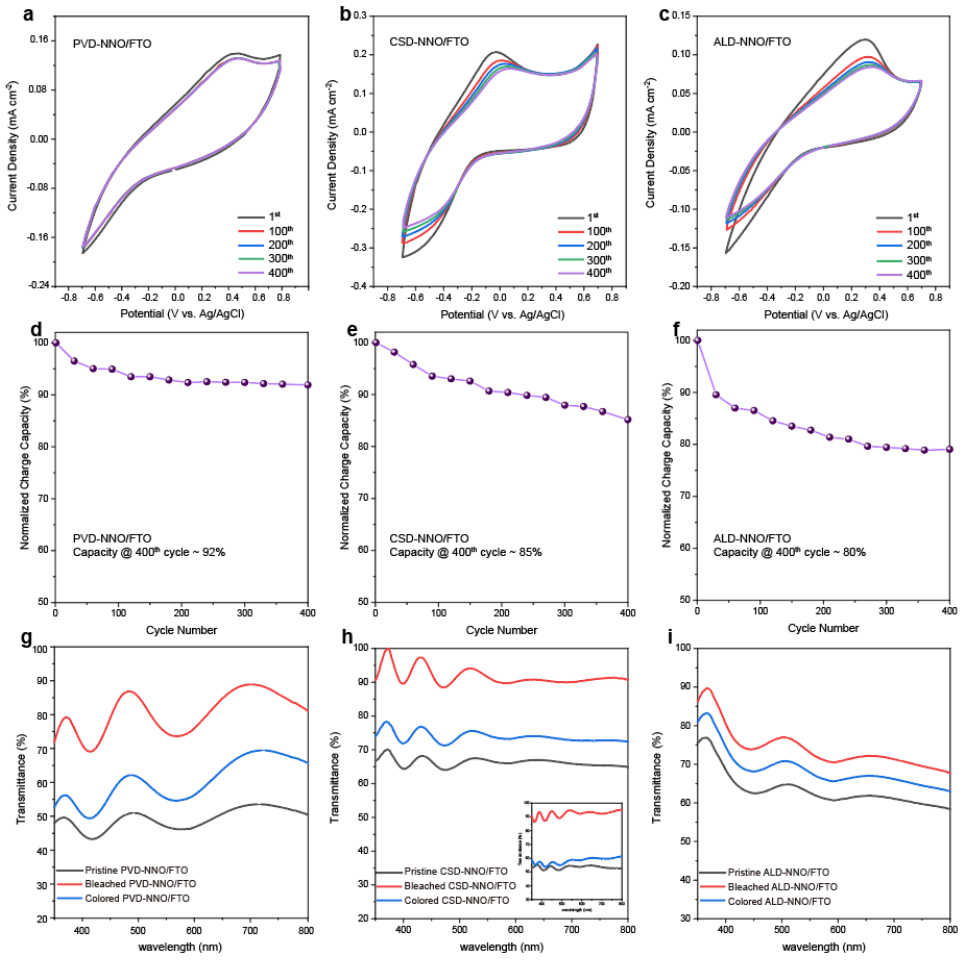


Figure 7. Polycrystalline NNO/FTO samples. (a-c) XRD patterns taken from ~200nm PVD-NNO/FTO, ~200nm CSD-NNO/FTO and ~50nm ALD-NNO/FTO and bare FTO glass substrate reference. Only one peak was identifiable in ~50nm ALD-NNO/FTO due to the thinner film. **(d-f)** Morphology of NNO films grown on FTO along with bare FTO substrates as reference for each case. NNO film prepared by PVD **(d-(ii))** or ALD **(f-(ii))** tends to follow the morphology of FTO **(d-(i) and (f-(i))**. The resulted polycrystalline PVD-NNO/FTO or ALD-NNO/FTO film is composed of randomly oriented small grains.



517

518 **Figure 8. Electrochromic properties of polycrystalline NNO/FTO thin films. (a-c)** CV stability test of
519 NNO/FTO thin films in pH=12 electrolyte. **(d-f)** The charge capacity maintained after 400 cycles was ~92%,
520 ~85% and ~80% for PVD-NNO/FTO, CSD-NNO/FTO and ALD-NNO/FTO, respectively. **(g-i)** UV-VIS
521 spectra of pristine, bleached (-0.7V vs. Ag/AgCl for 30s) and colored (+1V vs. Ag/AgCl for 60s) NNO/FTO
522 films with a thickness of ~50nm. The inset of **(h)** shows the UV-Vis spectra of 100nm CSD-NNO/FTO sample

at pristine, bleached (-0.7V vs. Ag/AgCl for 30s) and colored (+1V vs. Ag/AgCl for 180s) states, which shows a much better opaque/transparent contrast based on a comparable pristine transmittance with 50nm PVD-NNO/FTO **(g)** or 50nm ALD-NNO/FTO **(i).**